ARIJIT CHOWDHURY

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SKILLS

- **Languages:** C, C++, Python, JavaScript, Java
- **Frameworks & Libraries:** React, Next.js, Node.js, Express.js.
- **Databases:** MySQL, MongoDB.
- **DevOps & Tools:** Git, GitHub, Docker, Microservices, CI/CD, Agile Methodologies.
- **CS Core:** OOPS,SOLID,DBMS,Operating Systems,Computer Networks,DSA, REST API Design,Multithreading,Concurrency.

EXPERIENCE

1. Software Engineering Intern — Space Applications Centre (SAC) **Sep 2025 – Jan 2026**
Indian Space Research Organisation (ISRO) | Ahmedabad, India | [*Link*]

- Designed and trained an **Attention Residual U-Net** to correct systematic biases in numerical ocean model outputs using satellite altimeter observations as ground truth — achieving **52% SSIM and 32dB PSNR** (Arabian Sea) and reducing global **RMSE from 0.082m to 0.066m** on unseen test data.
- Formulated an end-to-end ML data pipeline benchmarking model predictions against 3 numerical ocean models across 2 regional zones, incorporating **bilinear interpolation, grid alignment, and anomaly isolation** to evaluate improvement over baseline forecasts across the North Indian Ocean.
- Engineered **hybrid loss functions** and AI-assisted Python pipelines, optimizing model training efficiency by **25%** across all regional zones.

PROJECTS

1. Tech Blogging App — AI-Powered Backend Platform **(Oct 2024 – Dec 2024)**
Technologies: Express, MongoDB, VectorDB, Redis, BullMQ, Gemini AI, ZOD, RAG, Auth | *[GitHub]*

- Designed a production-grade **RAG pipeline** by integrating **Google Gemini** embeddings with a **vector database, enabling semantic search** across blog content and returning **accurate, context-aware answers within 800ms**.
- **Built a distributed caching** and **async** job processing layer using **Redis** and **BullMQ**, reducing **database load** by **65–70%** via TTL-based cache invalidation, **cursor based pagination**, while delivering bulk newsletters at **5 jobs per second**.
- **Hardened API security** with passwordless **OTP** authentication, **HttpOnly JWT** session cookies, **role-based access control**, and **Zod** request validation — **eliminating 100%** of malformed payloads before reaching the controller.

2. NovaHTTP — Multithreaded C++ HTTP Server **(Jun 2026 - Aug 2026)**
Technologies: C++17, POSIX Sockets, Multithreading, Docker, CMake | *[GitHub]*

- Built a **multithreaded HTTP server** from scratch in C++ using raw **POSIX sockets**, featuring a custom request parser, response builder, and **segment-based router** supporting 6 REST endpoints with path parameters.
- Implemented a variable-size **thread pool** utilizing **mutex**, **condition variable** synchronization for safe concurrent connection handling, benchmarked system limits via **Apache Bench**, sustaining **280+ RPS** across **5,000 requests to 100 concurrent connections**, zero failures and **sub-180ms mean response time**.
- Containerized the build-to-run pipeline with Docker and CMake for one-command reproducible deployment, with thread-safe singleton logging and per-request execution-time tracking.

3. DSA Visualizer — Algorithm Visualization Platform **(Feb 2025 – Mar 2025)**
Technologies: React, JavaScript, Tailwind CSS, React Router | *[GitHub] [Live]*

- Engineered a non-blocking, step-driven visual execution engine leveraging **JavaScript Generator Functions** to handle complex asynchronous state tracking and real-time visualization across **7 core algorithms**.
- **Optimized rendering** with custom hooks, cutting re-renders by **40%** and delivering a consistently smooth **60FPS** interface.

4. AlgoSmith — C++ Template Library **(Feb 2026 – Mar 2026)**
Technologies: C++17, Templates, CMake, GoogleTest | *[GitHub] [Live]*

- Engineered a header-only **C++17 template library** implementing custom sequential and hierarchical data structures with strict **O(1) tail bounds** and **O(log n) search complexities**.
- Architected a decoupled structure separating **.hpp and .tpp files**, enforcing **Rule of Five** and **RAII** for leak-free memory, validated via **automated unit testing** with GoogleTest across **93/93 passed test cases**.

ACHIEVEMENTS

- Solved 600+ DSA problems on LeetCode and GFG, covering advanced data structures.
- Runner-Up, Annual College Coding Competition, 2024.
- Built an IoT-based smart fire detection system delivering. published in IoT Insights [View Chapter PDF].

EDUCATION

- B.Tech in CSE-AIML with CGPA: 8.2, MAKAUT University – 2026